Unit 4

Information Retrieval using NLP

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Information Retrieval (IR): Introduction

What is Information Retrieval (IR)?

- Information Retrieval (IR) is the process of obtaining relevant information from a large collection of unstructured or semi-structured data (e.g., text documents, web pages, or databases).
- It is widely used in **search engines, document retrieval systems, and AI-driven knowledge discovery**.

How IR Works

1. **Indexing** – Data is processed and stored in an optimized structure for retrieval.
2. **Query Processing** – A user inputs a query, which is interpreted by the system.
3. **Matching & Ranking** – Relevant documents are retrieved and ranked based on relevance.
4. **Presentation of Results** – The user receives a ranked list of results.

Applications of IR

- Search engines like **Google, Bing**
- Library databases (e.g., **PubMed, IEEE Xplore**)
- E-commerce search (e.g., **Amazon product search**)

Vector Space Model (VSM) in Information Retrieval

What is the Vector Space Model (VSM)?

- The Vector Space Model (VSM) is a *popular model used in IR* where documents and queries are represented as vectors in an **n-dimensional space**.
- **Each document** is treated as a **vector of terms** (words), and similarity between a query and documents is measured using techniques like **Cosine Similarity**.

Steps in VSM Approach

1. **Convert documents & queries into vectors** (using TF-IDF or Bag of Words).
2. **Compute similarity** between query and documents (using cosine similarity).
3. **Rank documents** based on similarity score.

Consider a Small Corpus of Documents

We have the following three documents in our dataset:

D1: "Machine learning is amazing."

D2: "Artificial intelligence and machine learning are related."

D3: "I love learning new things in artificial intelligence.“

And we have a **query**:

Q: "machine learning intelligence“

Our goal is to find which document is most relevant to the query using **VSM and Cosine Similarity**.

Create a Term-Document Matrix

Step 1: Extract Unique Words (Vocabulary)

Term	D1	D2	D3	Query
Machine	1	1	0	1
Learning	1	1	1	1
Is	1	0	0	0
Amazing	1	0	0	0
Artificial	0	1	1	1
Intelligence	0	1	1	1
And	0	1	0	0
Are	0	1	0	0
Related	0	1	0	0
I	0	0	1	0
Love	0	0	1	0
New	0	0	1	0
Things	0	0	1	0

Each **document** and **query** is now represented as a vector:

- **D1**: [1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0]
- **D2**: [1, 1, 0, 0, 1, 1, 1, 1, 1, 0, 0, 0, 0]
- **D3**: [0, 1, 0, 0, 1, 1, 0, 0, 0, 1, 1, 1, 1]

- **Q**: [1, 1, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0]

Compute Cosine Similarity

Cosine Similarity Formula:

$$\text{Similarity}(D, Q) = \frac{D \cdot Q}{\|D\| \times \|Q\|}$$

Where:

- $D \cdot Q$ = **Dot Product** of document and query vectors
- $\|D\|$ = **Magnitude (Norm) of Document Vector**
- $\|Q\|$ = **Magnitude of Query Vector**

Compute Dot Products

$$D1 \cdot Q = (1 \times 1) + (1 \times 1) + (0 \times 0) + (0 \times 0) + (0 \times 1) + (0 \times 1) = 2$$

$$D2 \cdot Q = (1 \times 1) + (1 \times 1) + (1 \times 1) + (1 \times 1) = 4$$

$$D3 \cdot Q = (0 \times 1) + (1 \times 1) + (1 \times 1) + (1 \times 1) = 3$$

- Compute Magnitudes

$$\|D1\| = \sqrt{1^2 + 1^2 + 1^2 + 1^2} = \sqrt{4} = 2$$

$$\|D2\| = \sqrt{1^2 + 1^2 + 1^2 + 1^2 + 1^2 + 1^2 + 1^2 + 1^2 + 1^2} = \sqrt{9} = 3$$

$$\|D3\| = \sqrt{0^2 + 1^2 + 1^2 + 1^2 + 1^2 + 1^2 + 1^2 + 1^2} = \sqrt{7} \approx 2.64$$

$$\|Q\| = \sqrt{1^2 + 1^2 + 1^2 + 1^2} = \sqrt{4} = 2$$

- Compute Cosine Similarities

$$\text{Similarity}(D1, Q) = \frac{2}{2 \times 2} = \frac{2}{4} = 0.5$$

$$\text{Similarity}(D2, Q) = \frac{4}{3 \times 2} = \frac{4}{6} = 0.67$$

$$\text{Similarity}(D3, Q) = \frac{3}{2.64 \times 2} = \frac{3}{5.28} \approx 0.57$$

Ranking the Documents

Based on **cosine similarity**, we rank the documents for the query **"machine learning intelligence"**:

Document	Cosine Similarity
D2: "Artificial intelligence and machine learning are related."	**0.67**
D3: "I love learning new things in artificial intelligence."	**0.57**
D1: "Machine learning is amazing."	**0.50**

D2 is the most relevant document, followed by D3 and D1.

Vector Space Model + Cosine Similarity

D1:
"Neural network optimization techniques improve deep learning performance in large scale data environments"

D2:
"Deep learning architectures such as convolutional neural networks are widely used in medical image analysis"

D3:
"Information retrieval systems apply vector space model and probabilistic ranking methods“

D4:
"Optimization algorithms including gradient descent improve neural network training efficiency"

D5:
"Medical diagnosis using machine learning and deep neural networks improves predictive accuracy"

Q:
"deep neural network optimization for medical diagnosis"

Perform the following steps:

1. Preprocess (lowercase, remove stopwords)
2. Construct full vocabulary
3. Compute:
    1. Term Frequency (TF)
    2. Document Frequency (DF)
    3. IDF using:

$$IDF(t) = \log_{10}\left(\frac{N}{df}\right)$$

4. Compute TF-IDF vectors
5. Normalize vectors
6. Compute cosine similarity between query and all documents
7. Rank documents

To Sum Up:-

- VSM converts **text into vectors** and measures similarity mathematically.
- Cosine Similarity is used to rank documents based on their relevance to a query.
- VSM is used in search engines, recommendation systems, and document retrieval.

Named Entity Recognition (NER):

What is Named Entity Recognition (NER)?

Named Entity Recognition (NER) is a NLP technique that **identifies and classifies named entities** in text into predefined categories such as:

- **Persons** (e.g., "Albert Einstein")
- **Organizations** (e.g., "NASA")
- **Locations** (e.g., "Himalayas")
- **Dates** (e.g., "21st June 2023")
- **Monetary Values** (e.g., "$1000")

How NER Works?

1. **Tokenization** – Splitting text into words.
2. **Part-of-Speech Tagging** – Identifying nouns, verbs, etc.
3. **Entity Recognition** – Assigning labels to detected entities.

Example of NER

- **Text:**
  "Elon Musk, the CEO of Tesla, announced a new AI project in California on January 10, 2024."

NER Output:

Entity	Category
Elon Musk	PERSON
Tesla	ORGANIZATION
AI	MISCELLANEOUS
California	LOCATION
January 10, 2024	DATE

Applications of NER

- **News Analytics** – Extract key people, places, and dates.
- **Chatbots** – Identify user intents.
- **Healthcare** – Extract disease names, drugs, and treatments.

System Building Process for Named Entity Recognition (NER)

- Named Entity Recognition (NER) is a crucial Natural Language Processing (NLP) task that **extracts and classifies named entities** such as **persons, locations, organizations, dates, and more** from text.
- The process of building an **NER system** involves several steps:

Data Collection & Preprocessing

To build an NER system, we first need a **dataset** containing **labelled entities**.

Datasets for NER:

- **CoNLL-2003** (Widely used, contains PERSON, LOCATION, ORGANIZATION, MISC categories)
- **OntoNotes 5.0** (More diverse dataset with additional entity types)
- **Custom dataset** (If working on specialized texts-domain specific)

Preprocessing Steps:

•**Tokenization** → Splitting text into words.
•**Lowercasing** → Standardizing words (optional).
•**Removing special characters & stop words** (unless they are meaningful in context).
•**Part-of-Speech (POS) Tagging** (to improve entity recognition).

Example:
Input Text:
"Elon Musk, CEO of Tesla, announced a new AI project in California on January 10, 2024.“

Preprocessed Tokens:
["Elon", "Musk", ",", "CEO", "of", "Tesla", ",", "announced", "a", "new", "AI", "project", "in", "California", "on", "January", "10", "2024", "."]

Choosing an NER Model

Several methods exist to develop an NER system:

Approach	Description	Examples
Rule-Based	Uses **predefined patterns** & dictionaries	**Regular expressions**, gazetteers
Statistical	Uses **probabilistic models** to identify entities	Hidden Markov Models (HMMs), Conditional Random Fields (CRFs)
Machine Learning	Uses **labeled datasets** to train classifiers	Support Vector Machines (SVMs), CRF-based models
Deep Learning	Uses **neural networks** for higher accuracy	LSTMs, BiLSTMs, Transformers (BERT, spaCy, Flair)

Example of a Rule-Based Approach:

Using **regular expressions** to detect dates:

```
import re

text = "Elon Musk was born on June 28, 1971."
date_pattern = r"\b(?:Jan|Feb|Mar|Apr|May|Jun|Jul|Aug|Sep|Oct|Nov|Dec)\s\d{1,2},\s\d{4}\b"
dates = re.findall(date_pattern, text)
print(dates)  # Output: ['June 28, 1971']
```

Example of a Deep Learning-Based Approach (Using spaCy):

```python
import spacy

nlp = spacy.load("en_core_web_sm")
text = "Apple Inc. was founded by Steve Jobs in California in 1976."
doc = nlp(text)

for ent in doc.ents:
    print(f"{ent.text} -> {ent.label_}")
```

Output:-

- Apple Inc. -> ORG
- Steve Jobs -> PERSON
- California -> GPE 1976 -> DATE

Training the NER Model

- If using a **machine learning or deep learning** approach, we must train the model with annotated data.
- **Feature Selection** (for ML models): Word embeddings, capitalization, POS tags, context.
- **Training a Deep Learning Model** (like BiLSTM-CRF or BERT-based models).
- **Fine-tuning** on domain-specific datasets (e.g., Yoga texts, Sanskrit manuscripts, medical drug discoveries).

Example of Training a Custom Model Using spaCy:

```python
import spacy
from spacy.training.example import Example

nlp = spacy.blank("en")  # Create a blank English model
ner = nlp.add_pipe("ner")  # Add Named Entity Recognizer to pipeline

# Add custom labels
ner.add_label("PHILOSOPHER")
ner.add_label("SPIRITUAL_CONCEPT")

# Create training data
train_data = [
    ("Patanjali wrote the Yoga Sutras.", {"entities": [(0, 9, "PHILOSOPHER"), (18, 29, "SPIRI
    ("Vedanta is an important philosophy.", {"entities": [(0, 7, "SPIRITUAL_CONCEPT")]}),
]
```

```python
# Training loop (simplified)
nlp.begin_training()
for text, annotations in train_data:
    example = Example.from_dict(nlp.make_doc(text), annotations)
    nlp.update([example])

# Test the model
doc = nlp("Who wrote the Yoga Sutras?")
for ent in doc.ents:
    print(ent.text, "->", ent.label_)
```

Evaluating the NER System (Entity Extraction Evaluation)

- Once we have built the NER system, we need to **evaluate its performance** using different metrics.
- Key Evaluation Metrics for NER

1. Precision
2. Recall
3. F1 Score

Relation Extraction (RE)

- Relation Extraction (RE) is a Natural Language Processing (NLP) technique used to identify and classify relationships between entities in text.
- It plays a crucial role in building knowledge graphs, information retrieval, and question-answering systems.

Types of Relation Extraction

1. **Supervised Learning-based RE:** Uses labeled datasets to train models for extracting relationships.
2. **Unsupervised Learning-based RE:** Identifies patterns in text without labeled data.
3. **Semi-Supervised & Distant Supervision RE:** Uses a small amount of labeled data or external knowledge bases to learn relationships.
4. **Rule-based RE:** Relies on predefined linguistic patterns and rules.

Example of Relation Extraction

Consider the sentence: 👉 *"Albert Einstein was born in Germany."*

Entities Identified:

- **Albert Einstein** (Person)
- **Germany** (Location)

Extracted Relation:

("Albert Einstein", **Born In**, "Germany")

Example:

"Google acquired DeepMind in 2014.“

- Entities: **Google** (Company), **DeepMind** (Company)
- Extracted Relation: ("Google", **Acquired**, "DeepMind")

Applications of Relation Extraction

- **Knowledge Graph Construction:** Extracts relationships for knowledge graphs like Google’s Knowledge Panel.
- **Biomedical NLP:** Identifies drug-disease relationships from research papers.
- **Event Extraction:** Helps in summarizing news articles by extracting event relationships.

Reference Resolution

- Reference resolution is the process of identifying **referring expressions** (like pronouns, noun phrases, demonstratives) and linking them to the correct **entity** mentioned in the discourse.
- It is crucial **for understanding context** in NLP applications such as **chatbots, machine translation, and summarization.**

Example:

Sentence

Riya submitted the paper. She was confident about it.

Here:

She → refers to Riya

it → refers to the paper

The system must correctly link:

(She → Riya)

(it → paper)

Types of Reference Resolution

Coreference Resolution: Determines when **two or more expressions** in text refer to the **same entity**.

"Einstein was a genius. He developed the theory of relativity."

“He" refers to **Einstein** (Coreference Resolution).

Anaphora Resolution: Identifies **pronouns** and resolves them to their **correct antecedents.**

"John bought a book. The book was about Artificial Intelligence."

"The book" refers to **"a book"** (Anaphora Resolution).

Cataphora Resolution: Resolves **forward references** where an entity appears later in the text

"Before she arrived, Maria had already planned the trip.“

"She" refers to **Maria** (Cataphora Resolution).

"Before he could say anything, John was interrupted."

- **Resolution:** "He" refers to **John**. (Cataphora Resolution).

"Although he was nervous, John gave a great speech."

- **Resolution:** "he" refers to **John**.

Both **Anaphora and Cataphora Resolution** are essential in NLP for tasks like **pronoun disambiguation, sentiment analysis, and text coherence improvement** in AI-based models

Coreference Resolution

- Coreference Resolution is the process of identifying and linking multiple expressions in a text that refer to the same entity.
- This **helps in understanding the context** of a sentence by associating pronouns or noun phrases with their corresponding subjects.

Types of Coreference Resolution

1. **Pronominal Coreference** – Resolving pronouns to their antecedents.
2. **Nominal Coreference** – Linking noun phrases that refer to the same entity.
3. **Identity Coreference** – Resolving multiple mentions of the same named entity.
4. **Bridging Anaphora** – Understanding implicit relationships between entities.

Examples of Coreference Resolution

Example 1 (Pronominal Coreference):

- ☞ *"Elon Musk founded SpaceX. He is also the CEO of Tesla.“*

- **Resolution:** "He" refers to **Elon Musk**.

Example 2 (Nominal Coreference):

- ☞ *"Barack Obama was the 44th U.S. President. The former president was known for his leadership.“*

- **Resolution:** "The former president" refers to **Barack Obama**.

Example 3 (Identity Coreference):

- *"Amazon acquired Whole Foods. The company made the acquisition for $13.7 billion."*
- **Resolution:** "The company" refers to **Amazon**.

Example 4 (Bridging Anaphora):

- *"I bought a book yesterday. The cover is beautiful."*

Resolution: "The cover" is related to **the book**, though "cover" is not explicitly mentioned earlier.

Cross-Lingual Information Retrieval (CLIR)

- Cross-Lingual Information Retrieval (CLIR) is the task of retrieving information written in one language when queries are issued in another language.
- It is useful for accessing global knowledge without language barriers.

CLIR Working

1. **Query Translation:** Converting the search query into the target language.
2. **Document Translation:** Translating indexed documents into the query language.
3. **Interlingual Representation:** Using a shared representation, such as embeddings, to match content across languages.

Examples of CLIR

Example 1 (Query Translation Method)

- **User Query:** "Weather in Paris" (English)
  Translation to French: "Météo à Paris"
  Retrieved Document: A French weather report about Paris is retrieved and either presented as is or translated back into English.

Example 2 (Document Translation Method)

- **Original Document:** A **Japanese** news article on AI development.
  User Query in English: "Latest AI research in Japan.“

Process: The system translates Japanese documents into English and ranks relevant results.

Example 3 (Interlingual Representation Method using Embeddings)

User Query in Spanish: "Mejorar la productividad“

English Document: "Boosting workplace efficiency“

Matching via Embeddings: The system recognizes "Mejorar la productividad" and "Boosting workplace efficiency" as similar concepts and retrieves the English document.

Applications of CLIR

- **Multilingual Search Engines** (e.g., Google Translate in search).
- **Academic Research Retrieval** (Accessing research papers in different languages).
- **Legal and Medical Information Access** (Providing critical information across languages).
- **Cross-Border Business Intelligence** (Understanding foreign market trends).

Both **Coreference Resolution and Cross-Lingual Information Retrieval** are crucial for **multilingual AI, document understanding, and global knowledge access**! 🌍